Institut Teknologi Sepuluh Nopember

STP ITS BLUEPRINT

DIAGNOSE · PLAN · ACTIVATE

The 100-day plan and the big plan for the **Science Techno Park, Institut Teknologi Sepuluh Nopember**

STP TEAM · UTM-ITS STELLAR 2026

Nur Hidayatul Alam · Biology

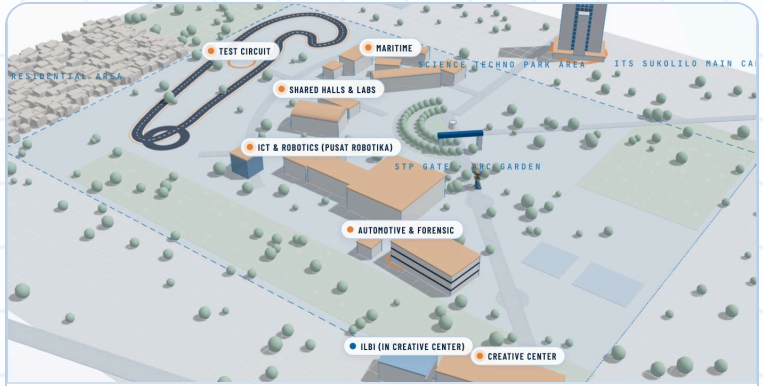
Rio Tria Hardhitya · Faculty of Electrical Engineering

Arta Kusuma Hernanda · Computer Engineering

Muhamad Syukron · Data Science



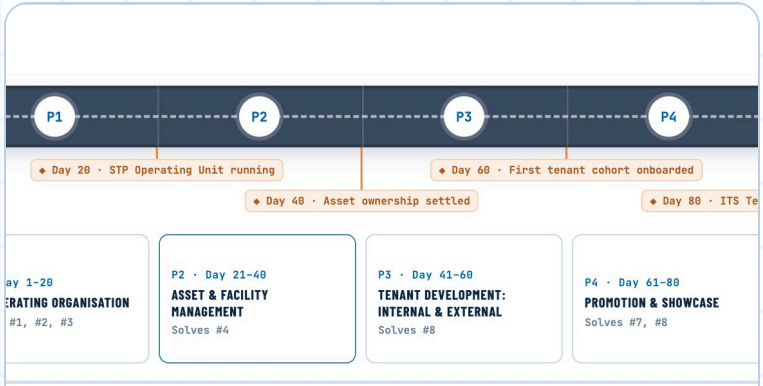
OUTLINE



01

NOW – EXISTING CONDITION

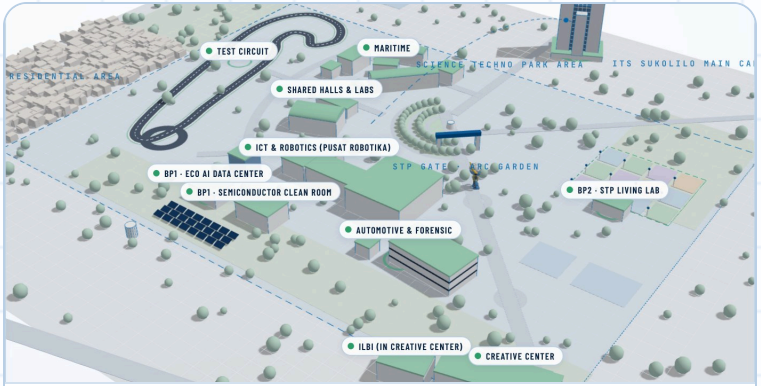
- STP ITS site map today
- Existing organisation (Bidang IV)
- Diagnosis: 8 issues



02

100-DAY PLAN

- Five priorities on one road (P1–P5)
- Quick wins & milestones
- Governance, resources, risks & policy



03

BIG PLAN

- Roadmap 2018 → 2045
- BP1 · Semiconductor & Eco AI Data Center
- BP2 · STP Living Lab

01

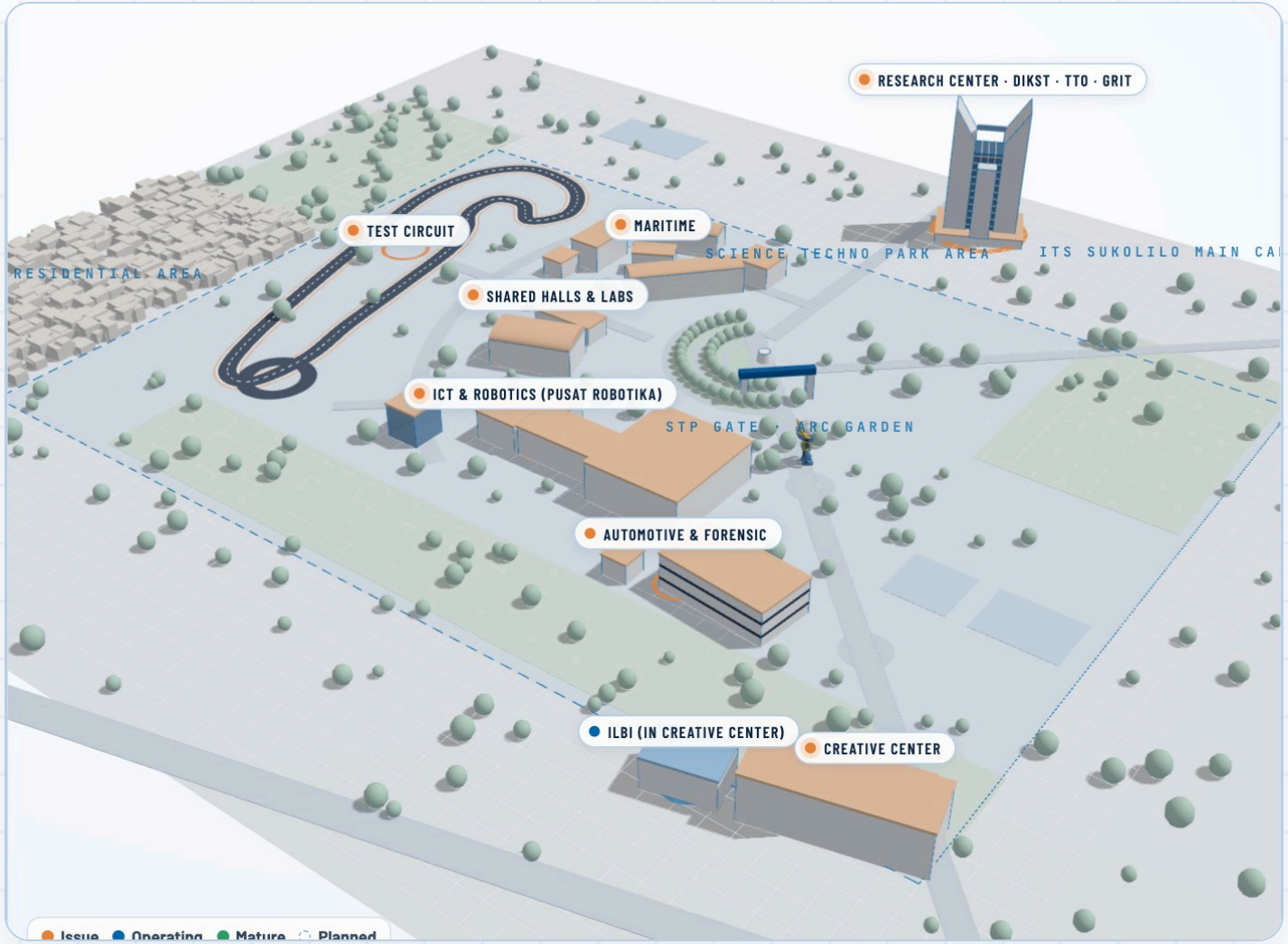
NOW

Where STP ITS stands today – the site, the organisation and the issues that block it.



01 · NOW

STP ITS TODAY



OVERVIEW

STP ITS sits under the Directorate of Innovation and Science & Technology Park (DIKST). It brings together the Technology Transfer Office (TTO), the Incubator & Innovative Business Services (ILBI) and four innovation clusters – Automotive, ICT & Robotics, Maritime and Creative Industry – each backed by Centres of Excellence (PUI). Its buildings and equipment were delivered through the HETI–ADB project.

✓ Rector Regulation No. 22/2026 on Science & Technology Park governance

✓ 4 clusters · 10+ PUI · HETI–ADB buildings

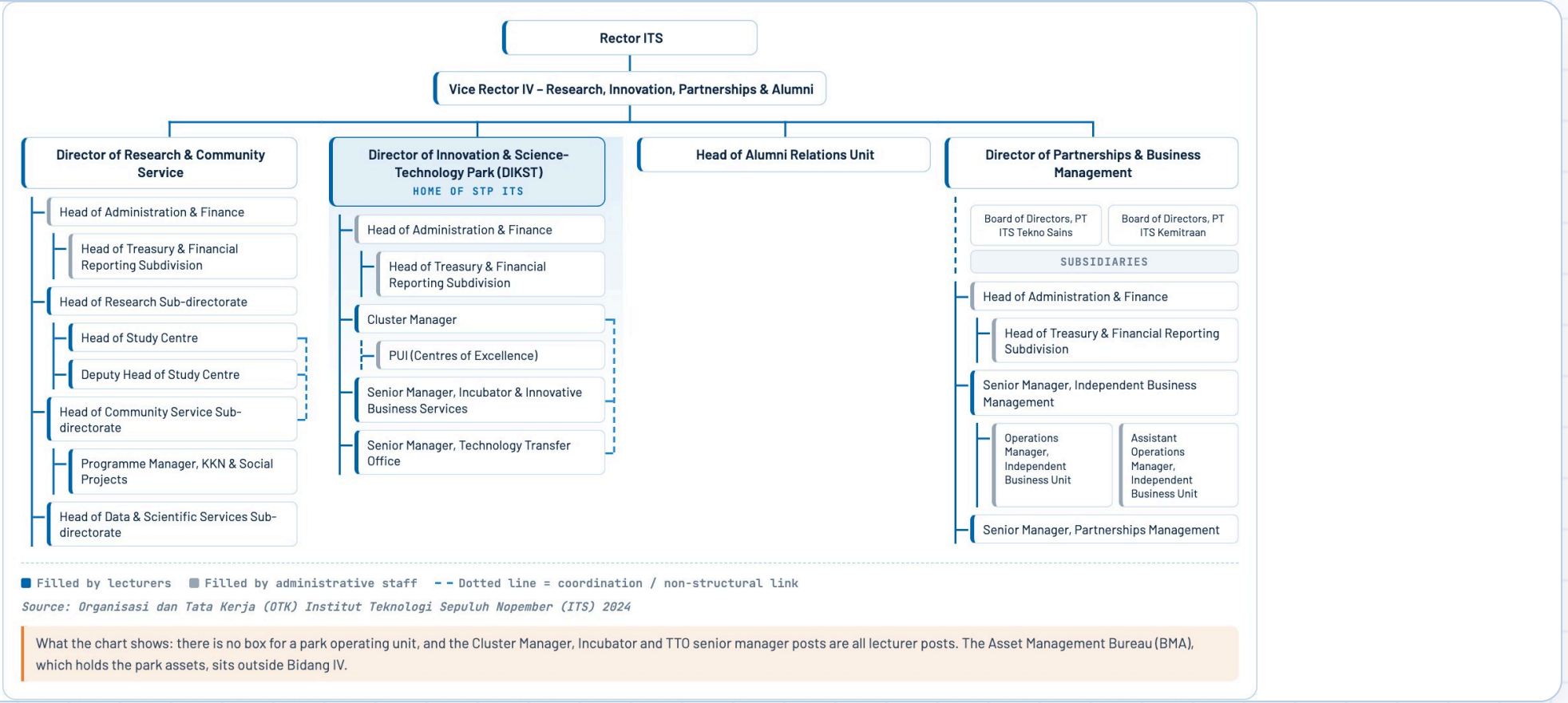
✓ 126 ILBI tenants (SIPENSI) · 130+ start-ups funded

TODAY

Hardware and regulation exist; the operating "software" – people, decision rights, systems and follow-up – is missing.

01 · NOW

EXISTING ORGANISATION – BIDANG IV



What the chart shows: there is no box for a park operating unit, and the Cluster Manager, Incubator and TTO senior manager posts are all lecturer posts. The Asset Management Bureau (BMA), which holds the park assets, sits outside Bidang IV.

01 · NOW

DIAGNOSIS – WHERE STP STANDS TODAY

The hardware (HETI–ADB buildings) and the rules (Rector Regulation 22/2026) are in place. What is missing is the software: people, decision rights, systems and follow-up.

#1 · P1

NOT YET AN INDEPENDENT ENTREPRENEURIAL ENTITY

IMPACT

STP still depends on ITS as the main entity for budget, staff and decisions; it cannot act or earn on its own.

ROOT CAUSE

STP is merged into DIKST, with no separate business unit, revenue model or delegated authority (Rector Regulation 22/2026 not yet implemented).

#2 · P1

SYSTEMS, ORGANISATION AND SOPS NOT YET MATURE

IMPACT

Processes depend on individuals; requests, approvals and hand-offs between units are slow and unclear.

ROOT CAUSE

No clear workflows, roles or decision paths – tenancy, facility, showcase and reporting SOPs still need to be written and aligned.

#3 · P1

MANAGERS ARE LECTURERS

IMPACT

Slow, inconsistent tenant services; overloaded staff.

ROOT CAUSE

No dedicated professional staff (park, tenant, facility, data).

#4 · P2

UNCLEAR ASSET OWNERSHIP

IMPACT

Ad-hoc use, safety and maintenance risk, unrecorded income.

ROOT CAUSE

No rule on who owns and who operates each asset – e.g. the circuit: BMA or STP?

#5 · P5

"SAME FACES AGAIN"

IMPACT

Few contributors, burnout risk, narrow ecosystem.

ROOT CAUSE

No incentives or open pathway for new contributors.

#6 · P5

NO DATA OR KPI TOOLS

IMPACT

No performance data; decisions by anecdote.

ROOT CAUSE

No booking, tenant register, lead CRM or KPI dashboard.

#7 · P4

SHOWCASE ONLY AT THE RESEARCH CENTER

IMPACT

Industry visits do not reach clusters or tenants.

ROOT CAUSE

One display in GRIT, no showcase in the cluster buildings and no regular rotation.

#8 · P3

FEW INDUSTRY PARTNERS AND TENANTS

IMPACT

Buildings are under-used, income is low and research rarely reaches industry.

ROOT CAUSE

No structured partnership programme, tenant offer or follow-up of industry leads.

02

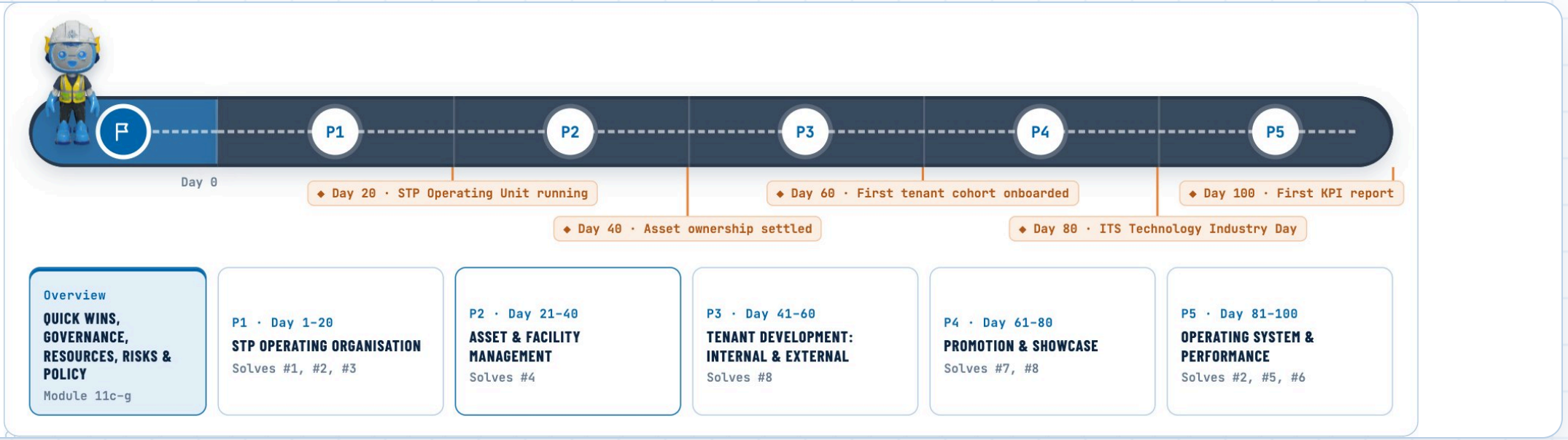
100-DAY PLAN

Five priorities run one after another on a single 100-day road – 20 days each. Every leg ends at a milestone.



02 · 100-DAY PLAN

ONE ROAD, FIVE PRIORITIES



P1 · DAY 1-20

STP OPERATING ORGANISATION

Solves #1, #2, #3

P2 · DAY 21-40

ASSET & FACILITY MANAGEMENT

Solves #4

P3 · DAY 41-60

TENANT DEVELOPMENT: INTERNAL & EXTERNAL

Solves #8

P4 · DAY 61-80

PROMOTION & SHOWCASE

Solves #7, #8

P5 · DAY 81-100

OPERATING SYSTEM & PERFORMANCE

Solves #2, #5, #6

P1 · Day 1–20 · Solves #1, #2, #3

STP OPERATING ORGANISATION

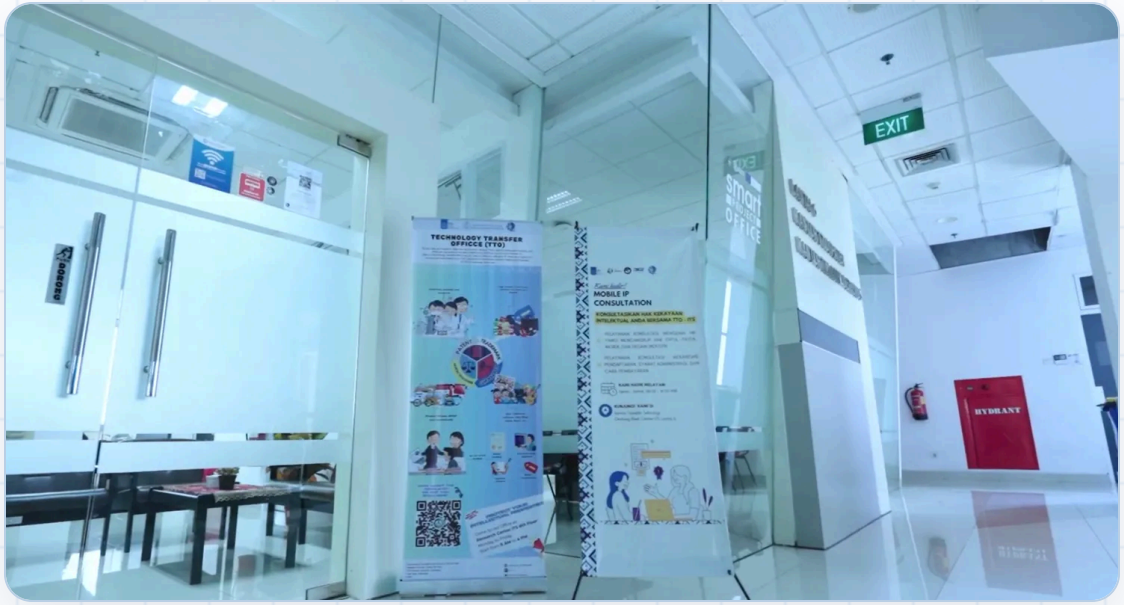
Without an owner and professional staff no other priority can be delivered.

OWNER

Director of DIKST · sponsor: Vice Rector IV

D1–7	Draft Rector decree for an STP Operating Unit (from Regulation 22/2026); job descriptions for 4 posts
D8–14	Assign or recruit a non-lecturer park manager, tenant, facility and data officers
D15–20	Unit operating; lecturers move to advisory roles in the clusters

◆ Day 20 · STP Operating Unit running



Technology Transfer Office – part of the operating core that the STP Operating Organisation coordinates

High impact · Low dependency · Visible · **Policy-enabling** · Pipeline-building · Revenue-enabling · Partner-backed

P2 · Day 21–40 · Solves #4

ASSET & FACILITY MANAGEMENT

Every asset needs one owner and one operator. Today nobody knows whether BMA or STP looks after some assets (the circuit is the clearest case), so maintenance, booking and income fall between the cracks.

OWNER

Director of DIKST + Head of BMA + Cluster Managers

D21–27	Register every park asset and facility (buildings, labs, circuit, equipment) with its current owner and operator
D28–34	BMA–STP agreement: an owner / operator matrix for every asset – BMA owns, STP operates – starting with the circuit case
D35–40	Facility maintenance SOP: preventive schedule, booking and safety rules, tariff card and a maintenance budget line

◆ Day 40 · Asset ownership settled



The Research Center tower and surrounding campus assets – each asset needs one owner and one operator

High impact · Low dependency · **Visible** · **Policy-enabling** · Pipeline-building · **Revenue-enabling** · **Partner-backed**

P3 · Day 41–60 · Solves #8

TENANT DEVELOPMENT: INTERNAL & EXTERNAL

Buildings only create value when they are occupied. The park needs internal tenants that grow from ITS (start-ups, spin-offs, student and lecturer ventures) and external tenants that bring demand, co-funding and income (industry, partners, government). Today the Maritime Cluster already hosts companies such as IDSurvey and IPERINDO – P3 turns such cases into a repeatable offer for every cluster.

OWNER

Senior Manager, Incubator & Innovative Business Services + Senior Manager, TTO + Cluster Managers

D41–47	Tenant offer & SOP: spaces in the 4 cluster buildings, two tracks – internal (incubation) and external (lease / R&D partnership) – with criteria, tariffs and a standard agreement
D48–54	Open call & outreach: internal start-ups and spin-offs via ILBI; external industry, partner and government tenants via the TTO, cluster networks and alumni
D55–60	First cohort onboarded: ≥ 5 internal and ≥ 2 external tenants with signed agreements, each linked to a cluster and a mentor

◆ Day 60 · First tenant cohort onboarded



Maritime Innovation Unit – external tenants already in the park (IDSurvey, Galangan Kapal Madura, Bintang Timur Samudera, IPERINDO)

High impact · Low dependency · Visible · Policy-enabling · Pipeline-building · Revenue-enabling · Partner-backed

P4 · Day 61–80 · Solves #7, #8

PROMOTION & SHOWCASE

Visible early credibility: innovations shown where they are made, and channels that reach industry, investors and future tenants.

OWNER

Gallery & communications coordinator (STP unit) + Cluster Managers + Senior Manager, TTO

D61–67	Gallery SOP: exhibit register, IP / disclosure clearance, rotation at least once per semester
D68–74	A showcase corner in each cluster building (Automotive, ICT & Robotics, Maritime, Creative) linked to GRIT as the hub; QR tech sheets
D75–80	Promotion relaunch: new STP website, YouTube, Instagram and LinkedIn content calendar; ITS Technology Industry Day (10 technologies × 20 companies)

◆ Day 80 · ITS Technology Industry Day



ITS innovations shown to the public (i-CAR) – the model for showcase corners and promotion

High impact · Low dependency · Visible · Policy-enabling · Pipeline-building · Revenue-enabling · Partner-backed

P5 · Day 81–100 · Solves #2, #5, #6

OPERATING SYSTEM & PERFORMANCE

Without shared tools, data and an open circle of contributors, the park keeps running on a few people and decisions stay anecdotal. P5 turns the units, SOPs and assets from P1–P4 into one measurable system.

OWNER

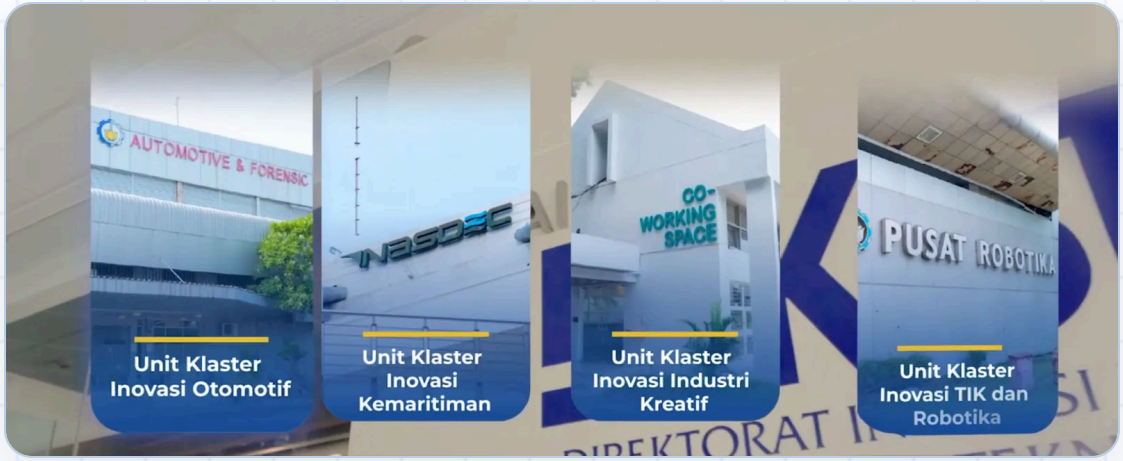
STP Operating Unit (data officer) · reporting to the Director of DIKST

D81–87	One operating system: online facility booking, tenant & asset register, lead CRM and document hub – start low-code, one source of truth
D88–94	KPI dashboard live (occupancy, utilisation, tenants, leads, income, showcase visits); open call for "STP Fellows" with workload credit or honorarium
D95–100	First quarterly KPI report to Vice Rector IV with improvement actions; ≥ 10 new contributors beyond the usual names

◆ Day 100 · First KPI report



The STP operating core: Technology Transfer Office and Incubator & Innovative Business Services



Four cluster units that feed one shared system and KPI dashboard

High impact · **Low dependency** · Visible · **Policy-enabling** · **Pipeline-building** · Revenue-enabling · Partner-backed

02 · 100-DAY PLAN

QUICK-WIN SELECTION

PRIORITY	HIGH IMPACT	LOW DEPENDENCY	VISIBLE	POLICY-ENABLING	PIPELINE-BUILDING	REVENUE-ENABLING	PARTNER-BACKED
P1 · STP Operating Organisation	✓			✓			
P2 · Asset & Facility Management	✓		✓	✓		✓	✓
P3 · Tenant Development: Internal & External	✓		✓		✓	✓	✓
P4 · Promotion & Showcase	✓	✓	✓		✓		✓
P5 · Operating System & Performance		✓		✓	✓		

QUICK WIN · DAY 40

ASSET OWNERSHIP SETTLED

Every park asset and facility has one owner and one operator in a signed BMA–STP agreement, so a case like the circuit – BMA or STP? – no longer happens. Maintenance, booking and tariffs follow from it.

QUICK WIN · DAY 80

ITS TECHNOLOGY INDUSTRY DAY

Showcase of ITS's top 10 technologies to 20 companies, starting at GRIT and continuing to the cluster showcase corners, ending with MoU/NDA signing.

02 · 100-DAY PLAN

GOVERNANCE & ACCOUNTABILITY

WORKSTREAM	EXECUTIVE SPONSOR	PROJECT LEAD	DECISION FORUM	REPORTING
STP governance & staffing	Vice Rector IV	Director of DIKST	Rector's leadership meeting	Monthly (100 days)
Assets & infrastructure	Vice Rector IV + Vice Rector overseeing BMA	Head of BMA + Park Manager + Automotive Cluster Manager	Park Asset Committee	Monthly
Tenants & incubation	Director of DIKST	Senior Manager, ILBI	Tenancy & Incubation Committee	Monthly
Showcase & industry engagement	Director of DIKST	Gallery coordinator + Senior Manager, TTO	Gallery Curation Committee	Per event + quarterly
Data & KPIs	Director of DIKST	STP data officer	DIKST performance review	Quarterly

02 · 100-DAY PLAN

RESOURCE PLAN

RESOURCE	AVAILABLE	GAP	ACTION
People	Lecturer-managers; DIKST admin staff	No dedicated park professionals	4 posts (re-assign staff first) + STP Fellows
Operating budget	DIKST unit budget	No line for facility running costs and events	Budget line + reinvest service income
Systems & data	Spreadsheets, SIPENSI	No booking, tenant register, CRM, dashboard	Start low-code; integrate later
Facilities	4 cluster buildings, circuit, GRIT	Under-used; no access rules	SHARE first – no new building in 100 days
Network	Cluster partners, alumni, KSTI 2026 host role	Leads not followed up	Industry Day + CRM + 5-day follow-up rule
Digital channels	STP website, DIKST YouTube, Instagram	Unattractive, irregular content	New web (this blueprint) + monthly content calendar

02 · 100-DAY PLAN

RISKS & POLICY HANDOVER

RISK	MITIGATION	OWNER
Decree and staffing approved late	Derive directly from Regulation 22/2026; start with staff re-assignment	Director of DIKST
BMA and STP disagree on who owns or operates an asset	Joint forum at vice-rector level; one owner / operator matrix for all assets, starting with the circuit	Vice Rector IV
Few first tenants	Prioritise ILBI alumni (126 SIPENSI tenants) and cluster partners	Senior Manager, ILBI
Industry Day without follow-up	Lead CRM, named owner per lead, 5-working-day rule	Senior Manager, TTO
Still the same few contributors	Workload credit / honorarium; open STP Fellows call	Director of DIKST
Big projects become a cost burden	Run them as two big projects with phases, partners (grant / co-funding / PPP) and a maintenance budget from day one	Director of DIKST

POLICY HANDOVER

1. Rector decree: STP Operating Unit and job descriptions (derived from Regulation 22/2026)
2. BMA–STP asset agreement: owner / operator matrix for every asset (incl. the circuit)
3. Facility maintenance SOP: preventive schedule, booking, safety rules and tariff card
4. Tenancy SOP: enquiry → application → screening → agreement → handover → monitoring → exit
5. Incubation SOP: admission, milestones, graduation / exit
6. Gallery SOP for GRIT and the cluster showcase corners: curation, IP clearance, rotation at least once per semester, visits and lead follow-up
7. STP Fellows incentive policy
8. STP KPI framework and reporting cycle
9. Communication SOP: content calendar and owners for the website, YouTube, Instagram and LinkedIn

03

BIG PLAN

New facilities are grouped into two big projects that follow the 100 days. The 100 days do not build them – they put in place the governance, asset rules, maintenance SOPs and promotion that a big project needs to succeed.



03 · BIG PLAN

ROADMAP & THE PARK IN THE BIG PLAN

2018
INITIATION · 0-3 YRS
Founding, basic labs, TTO and ILBI still separate

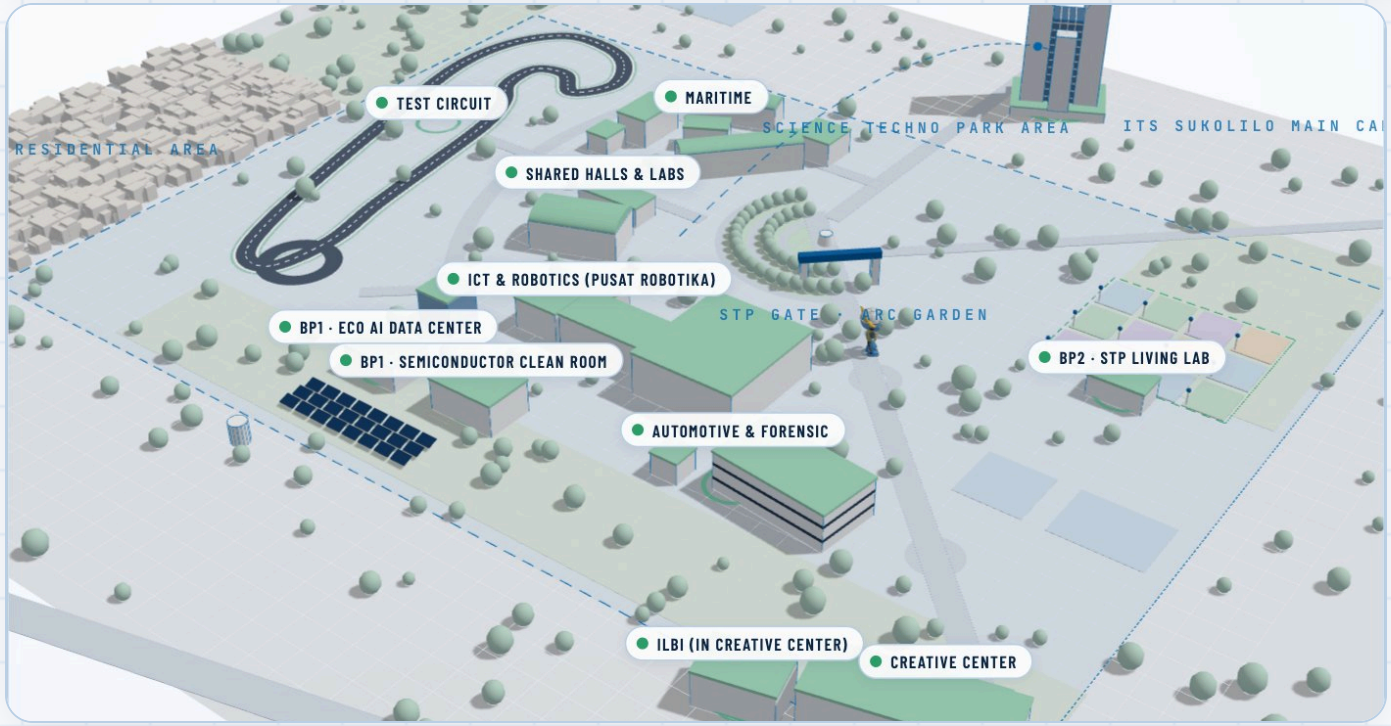
2020
INCUBATION · 3-7 YRS
DIKST formed: TTO + ILBI + 4 clusters + PUI

2026
WE ARE HERE
Rector Regulation 22/2026 · HETI–ADB buildings · this 100-day plan

2027
GROWTH · 7-12 YRS
Downstreaming: teaching factories, grants as projects, pilot products

2030
SELF-SUSTAINING · 12-20 YRS
Teaching factory + start-ups at the core, flagship products, service businesses

2045
REVENUE & IMPACT
Knowledge-based economy and regional competitiveness



BP1 · ICT & ROBOTICS CLUSTER

SEMICONDUCTOR & ECO AI DATA CENTER AREA

A clean room for semiconductor learning and prototyping, and a renewable, low-water Eco AI Data Center – one technology area serving chip, AI and digital research and tenants.

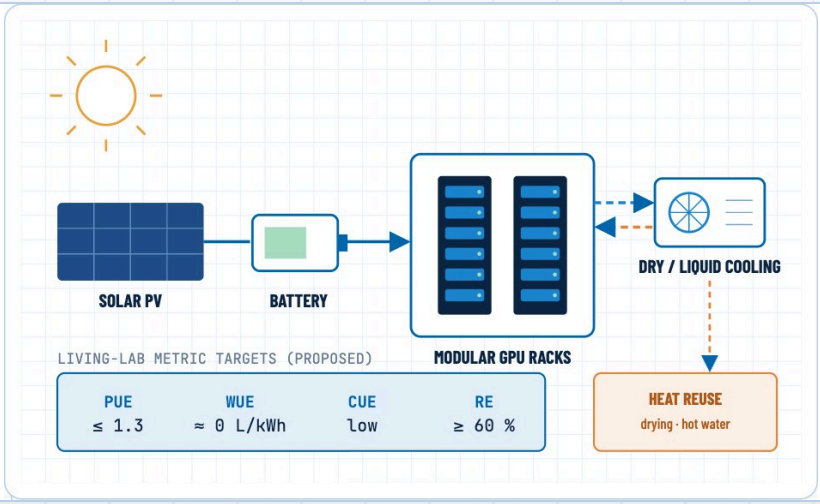
BP2 · PARK-WIDE, WITH THE TEST CIRCUIT

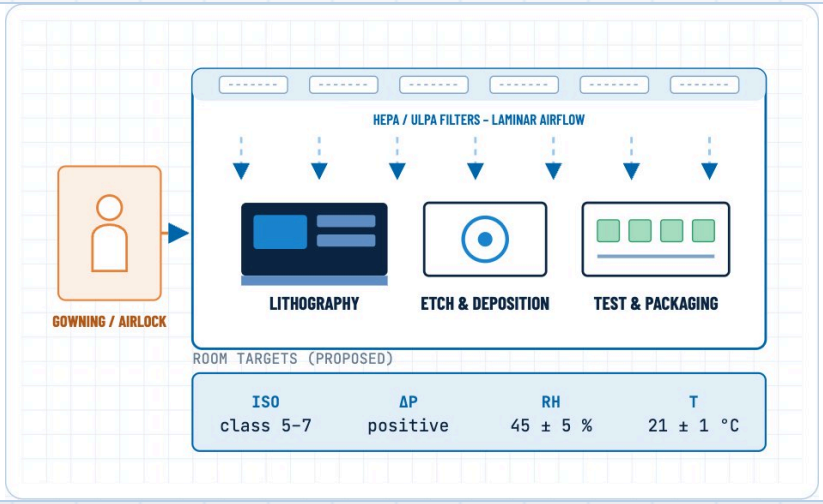
STP LIVING LAB

An open, real-world testbed where clusters and tenants pilot technologies with real users and shared data.

03 · BIG PLAN · BP1 · ICT & ROBOTICS CLUSTER

SEMICONDUCTOR & ECO AI DATA CENTER





SOLAR PV + BATTERY

Rooftop / ground-mount solar with battery storage; grid as backup

LOW-WATER COOLING

Direct-to-chip liquid or dry cooling, free cooling at night; no evaporative towers

MODULAR GPU RACKS

Start with 1–2 racks for AI research and tenant compute services

LIVE METRICS

PUE, WUE and CUE dashboards as open data for research and teaching

FILTERED LAMINAR AIRFLOW

HEPA / ULPA ceiling filters and positive pressure keep the room at ISO class 5–7

CORE PROCESS TOOLS

Lithography, etch & deposition and test & packaging bays for teaching and prototyping

GOWNING & AIRLOCK

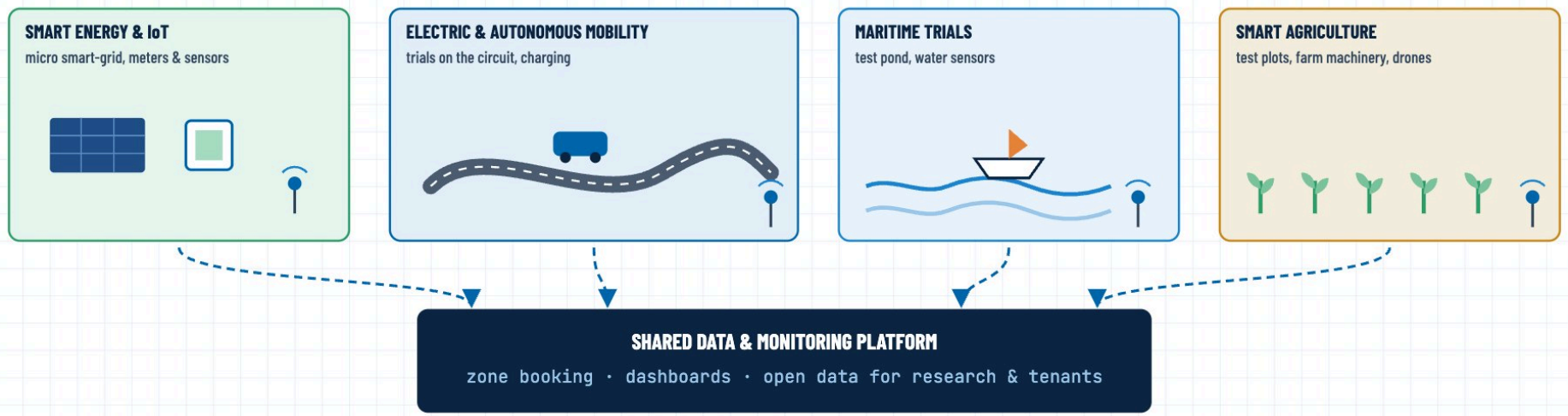
Controlled entry, training and access rules for students, researchers and tenants

ENVIRONMENT MONITORING

Particle count, pressure, humidity and temperature logged live – the base of the maintenance plan

03 · BIG PLAN · BP2

STP LIVING LAB



SMART ENERGY & IOT

A micro smart-grid with solar, storage, meters and sensors across the park

ELECTRIC & AUTONOMOUS MOBILITY

EV and autonomous-vehicle trials on the test circuit, with charging points

MARITIME TRIALS

Test pond and water sensors for the Maritime Cluster and its partners

SMART AGRICULTURE

Test plots for farm machinery, drones and sensor-based farming with users

SHARED DATA PLATFORM

Zone booking, dashboards and open data so clusters and tenants learn from every pilot

ITS
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Sepuluh Nopember

THANK YOU

Explore the interactive blueprint – 3D site map, 100-day road and big plan.

bit.ly/stpbp

STP team · UTM–ITS STELLAR 2026

